**Technical Assessment**

**Aeronautical Data Engineer Intern – CopilOPS**

**Objective**

The purpose of this assessment is to evaluate your ability to extract, structure, validate, visualize and maintain aeronautical information from an official State AIP.

The final objective is to demonstrate how aeronautical information could be transformed into structured operational data usable by an aviation operational decision-support platform.

This assessment is focused on:

* data extraction;
* backend/data engineering;
* geospatial data;
* automation;
* data validation;
* AIRAC lifecycle management;
* operational usefulness.

The visual design of the interface is secondary to the quality of the data and the architecture.

**1. Official data source**

Use only the official Belgium & Luxembourg AIP published by skeyes as the primary aeronautical information source.

The source must be:

* official;
* publicly accessible;
* free of charge.

Do not use commercial aviation databases or unofficial websites as substitutes for AIP information.

Unofficial sources may only be used for technical research, not as the authoritative data source.

**2. Airports to analyze**

The assessment must cover the following three airports:

* EBBR — Brussels Airport
* EBLG — Liège Airport
* EBCI — Brussels South Charleroi Airport

In addition, relevant ENR information affecting these airports and Belgian airspace must be extracted.

**3. Required aeronautical data**

Extract and structure the information that would be useful to a Flight Dispatcher, OCC or Operations Controller.

At minimum, include:

Aerodrome information

* ICAO code
* airport name
* coordinates
* elevation
* magnetic variation
* operating hours
* airport operator
* airport contact details
* AIS contacts
* ATS contacts
* handling contacts where published
* customs and immigration information
* fuel availability
* rescue and firefighting category
* operational restrictions
* PPR requirements where applicable

**Runway information**

For each runway:

* runway designator
* coordinates
* dimensions
* surface
* strength / PCN / PCR where applicable
* TORA
* TODA
* ASDA
* LDA
* displaced thresholds
* slope where available
* relevant lighting information

**Navigation information**

Extract:

* VOR
* DME
* NDB
* ILS
* other published radio navigation facilities

For each facility, provide where available:

* identifier
* type
* frequency
* coordinates
* associated airport or airspace

**Waypoints**

Extract significant published waypoints including:

* identifier
* latitude
* longitude

Coordinates must be converted into a format suitable for geospatial processing.

**ATS routes / airways**

Extract the Belgian ATS route network where available.

Include:

* route designator
* waypoint sequence
* direction restrictions
* lower limit
* upper limit
* relevant navigation specification

Where applicable, identify navigation requirements such as:

* RNAV
* RNAV 1
* RNP
* RNP 5
* other published navigation specifications

**Airspace**

Extract and visualize relevant:

* FIR
* UIR
* CTA
* TMA
* CTR

Include:

* designation
* horizontal geometry
* lower limit
* upper limit
* controlling authority where available

**Special-use and temporary airspace**

Include relevant:

* prohibited areas;
* restricted areas;
* danger areas;
* TRA;
* TSA;
* temporary reserved or restricted areas;

* temporary airspace published through AIP Supplement.

The system should distinguish between permanent and temporary information.

**Procedures**

It is not necessary to reproduce complete instrument procedures.

Extract useful procedure metadata such as:

* SID availability
* STAR availability
* approach types
* RNAV/RNP requirements
* procedure-related navigation specifications

**Operational information**

Identify other information contained in the AIP that could reasonably affect daily operations or flight planning.

Examples:

* airport restrictions;
* noise restrictions;
* curfews;
* runway limitations;
* special procedures;
* local operational requirements;
* significant ground limitations;
* special handling requirements.

You are expected to determine what information is operationally relevant rather than simply copying AIP tables.

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**4. Interactive HTML map**

Create an interactive HTML-based map presenting the extracted information.

The application may run locally.

It should contain selectable layers such as:

Airports

* EBBR
* EBLG
* EBCI

Navigation

* waypoints
* VOR
* DME
* NDB
* other navaids

Routes

* ATS routes / airways

Airspace

* FIR
* CTA
* TMA
* CTR

Special-use airspace

* prohibited areas
* restricted areas
* danger areas
* temporary areas

Selecting an element should display the relevant operational information.

Example:

Waypoint: XXXXX
Coordinates:
50.1234° N
004.5678° E
Associated routes:
Lxxx
Uxxx
Navigation requirement:
RNP 5
Source:
Belgium AIP ENR xx
Effective date:
YYYY-MM-DD

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**5. Airport operational view**

Each of the three airports should also have an operational summary.

Example structure:

EBBR — BRUSSELS
Runways
01/19
07L/25R
07R/25L
Operating hours
H24
RFFS
Category XX
Customs
Available
Navigation
ILS
RNAV 1
RNP APCH
Contacts
Airport operator
AIS
ATS
Handling
Operational restrictions
...
Current AIP effective date
...

The objective is to transform AIP information into something immediately understandable by an operational user.

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**6. Data provenance**

Every extracted value must remain traceable to its original source.

The database should retain information such as:

```
{
  "value": "3211",
```

```
  "unit": "M",
  "source": "Belgium AIP",
  "section": "EBBR AD 2.13",
  "effective_date": "YYYY-MM-DD",
  "retrieved_at": "YYYY-MM-DDTHH:MM:SSZ",
  "source_url": "...",
  "parser_version": "1.0"
}
```

We must always be able to determine:

* where the data came from;
* when it became effective;
* when it was retrieved;
* which parser generated it.

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**7. AIRAC update strategy**

A major part of this assessment is the design of an automated update mechanism.

Aeronautical information changes according to the AIRAC cycle, normally every 28 days.

Design a strategy allowing the database to be updated with minimal manual intervention.

The expected concept should cover:

Official AIP source
↓
Detect new AIRAC version
↓
Download / retrieve new publication
↓
Parse data
↓
Normalize into common schema
↓
Validation
↓
Compare CURRENT vs NEW
↓
Detect changes
↓
Review / automated quality checks

↓
Production database

The system should detect at least:

* NEW
* MODIFIED
* DELETED
* UNCHANGED

records.

Example:

EBBR
Runway 25L
Previous AIRAC:
LDA = 3211 m
New AIRAC:
LDA = 3151 m
CHANGE DETECTED
Field:
LDA
Previous:
3211
New:
3151
Source:
EBBR AD 2.13

Do not simply replace the complete database every 28 days.

The objective is to identify and manage changes.

**8. Temporary information**

Explain how your architecture would differentiate:

Permanent information

AIP

Medium-term temporary information

AIP Supplement

Short-term operational information

**NOTAM**

The full NOTAM implementation is not part of this assessment.

However, your architecture should demonstrate how NOTAM information could later be integrated without mixing it with permanent AIP data.



**9. Data validation**

Describe how you would detect problems such as:

* invalid coordinates;
* missing runway data;
* duplicated waypoints;
* incorrect units;
* broken references between routes and waypoints;
* unexpected HTML/XML structure changes;
* incomplete extraction;
* disappearance of previously existing data;
* unexpected changes between AIRAC cycles.

The system should not silently accept corrupted data.

**10. Engineering considerations**

Explain how your solution handles situations such as:

* the AIP publisher changes an HTML table;
* an AIP section changes structure;
* a coordinate cannot be parsed;
* an airport runway disappears between AIRAC cycles;
* an operating-hour field changes from H24 to a textual restriction;
* an entire AIP section fails to parse;
* the source website is temporarily unavailable.

We are interested in how you design a maintainable system, not only whether the first extraction succeeds.

**11. Deliverables**

Provide:

1. Source code
2. README
3. Structured extracted dataset
4. Interactive HTML map
5. Database/data schema
6. AIRAC update strategy
7. Change-detection concept
8. Data-validation approach
9. List of data-quality issues or ambiguities identified in the AIP
10. Maximum two-page technical explanation of your architecture

The project should be runnable by another developer using the instructions provided in the README.

**12. Technical freedom**

Use the tools mentioned or use tools you consider appropriate and explain the reasoning behind important architectural decisions.

**13. Use of AI**

AI tools may be used as development assistants.

However:

* the extraction pipeline must remain reproducible;
* critical aeronautical data must be deterministically validated;
* an LLM should not be the sole mechanism used to determine or generate aeronautical values;
* you must understand and be able to explain the submitted code and architecture.

**14. Scope and time**

This assessment is intentionally broad.

We do not expect the entire Belgium AIP to be completely converted into a production-ready database.

Focus on demonstrating a scalable architecture.

Recommended maximum effort:

8–12 hours

Priority should be given to:

1. EBBR, EBLG and EBCI;
2. waypoints and navaids;
3. ATS routes;
4. airspace;
5. operationally relevant AIP information;
6. interactive map;
7. AIRAC automation;
8. data provenance and validation.

Partial implementation with a strong scalable architecture is preferable to a large amount of hard-coded data.

**15. Evaluation criteria**

The assessment will primarily be evaluated on:

| Area | Weight |
|---|---|
| Data extraction and parsing | 20% |
| AIRAC automation and change detection | 20% |
| Data modelling | 15% |
| Operational relevance | 15% |
| Validation and error handling | 10% |
| Geospatial/map implementation | 10% |
| Code quality and documentation | 5% |
| Architecture and initiative | 5% |

The purpose of this assessment is not to test prior aviation knowledge.

We are primarily assessing your ability to understand a complex structured information source and transform it into a reliable, maintainable and scalable aeronautical data pipeline